Project Assignment

Comparative Study of Traditional Machine Learning and

Transformer Models for Spam Classification

Objective

You are required to design and implement two different approaches to classify emails into Spam and Ham. You will compare a traditional machine learning pipeline with a transformer-based deep learning model using DistilBERT. The goal is to understand the difference between classical feature engineering approaches and contextual representation learning.

Dataset

You must use the following dataset:

https://www.kaggle.com/datasets/ashfakyeafi/spam-email-classification

Before building models, you are expected to explore the dataset, understand class distribution, identify inconsistencies or preprocessing needs, and justify your data split strategy.

Part A – Traditional Machine Learning

Implement a classical NLP pipeline including text preprocessing, TF-IDF feature representation, and at least two classifiers: Logistic Regression and Naive Bayes. Evaluation must include Accuracy, Precision, Recall, F1-score, and Confusion Matrix. All preprocessing and vectorization decisions must be justified.

Part B – Transformer-Based Model

Implement a spam classifier using DistilBERT from Hugging Face with PyTorch only. You must implement your own PyTorch training loop and fine-tune the model for classification. Do not use high-level Trainer APIs. You are responsible for deciding learning rate, batch size, number of epochs, layer freezing strategy, and any regularization methods. Evaluation metrics must be identical to Part A.

Visualization Requirements

All comparisons must be visualized using Matplotlib and Seaborn. You must present:

- Metric comparison across models
- Confusion matrices
- Training behavior of the transformer model

- Visual comparison of computational cost (e.g., training time)

All plots must be clearly labeled and interpretable.

Deliverables

You must submit:

1. Well-structured source code
2. A technical report (3–5 pages) including methodology, design decisions, experimental setup, results, and comparative discussion
3. All visualizations
4. Clear explanation of limitations